



nextLeap



Code



Input



Output

```
1  import pandas as pd
2
3  # Sample dataset
4
5  data = {
6
7      "Name": ["Ali", "Sara", "John", "Sara",
8              None],
9
10     "Age": [20, 22, 21, None, 23],
11
12     "Department": ["IT", "HR", "IT", "HR",
13                   "IT"],
14
15     "Salary": [30000, 35000, 32000, 36000,
```

```
None]
```

```
14
```

```
15 }
```

```
16
```

```
17 df = pd.DataFrame(data)
```

```
18
```

```
19 print("Original Data:")
```

```
20
```

```
21 print(df)
```

```
22
```

```
23 # Clean missing values
```

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Code



Input



Output

```
22
23 # Clean missing values
24
25 ✓ df = df.fillna({
26     "Name": "Unknown",
27     "Age": df["Age"].mean(),
28     "Salary": df["Salary"].mean()
29 })
30
31
32
33
34
35 print("\nCleaned Data:")
36
37 print(df)
```

```
37 print(df)
38
39 # Filter data
40
41 it_employees = df[df["Department"] ==
42 "IT"]
43
44 print("\nIT Employees:")
45
46 print(it_employees)
```

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**Code****Input****Output**

32

33 })

34

35 print("\nCleaned Data:")

36

37 print(df)

38

39 # Filter data

40

41 it_employees = df[df["Department"] ==
 "IT"]

42

43 print("\nIT Employees:")

44

45 print(it_employees)

46

```
47  # Grouping and aggregation
48
49  grouped = df.groupby("Department")
      ["Salary"].mean()
50
51  print("\nAverage Salary by Department:")
52  print(grouped)
```

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Input



Output

Original Data:

	Name	Age	Department	Salary
0	Ali	20.0	IT	30000.0
1	Sara	22.0	HR	35000.0
2	John	21.0	IT	32000.0
3	Sara	NaN	HR	36000.0
4	None	23.0	IT	NaN

Cleaned Data:

	Name	Age	Department	Salary
0	Ali	20.0	IT	30000.0
1	Sara	22.0	HR	35000.0
2	John	21.0	IT	32000.0
3	Sara	21.5	HR	36000.0

```
4 Unknown 23.0      IT 33250.0
```

IT Employees:

	Name	Age	Department	Salary
0	Ali	20.0	IT	30000.0
2	John	21.0	IT	32000.0
4	Unknown	23.0	IT	33250.0

Average Salary by Department:

Department
HR 35500.0

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Output

0	Ali	20.0	IT	30000.0
1	Sara	22.0	HR	35000.0
2	John	21.0	IT	32000.0
3	Sara	NaN	HR	36000.0
4	None	23.0	IT	NaN

Cleaned Data:

	Name	Age	Department	Salary
0	Ali	20.0	IT	30000.0
1	Sara	22.0	HR	35000.0
2	John	21.0	IT	32000.0
3	Sara	21.5	HR	36000.0
4	Unknown	23.0	IT	33250.0

IT Employees:

```
      Name  Age Department  Salary
0    Ali  20.0         IT  30000.0
2   John  21.0         IT  32000.0
4 Unknown 23.0         IT  33250.0
```

Average Salary by Department:

```
Department
```

```
HR    35500.0
```

```
IT    31750.0
```

```
Name: Salary, dtype: float64
```

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